Indian J Surg (December 2015) 77(Suppl 2):S708–S711
DOI 10.1007/s12262-013-0937-x

'Single-Sitting' Laparoscopic Cholecystectomy and Endoscopic Removal of Common Bile Duct Stone for Cholelithiasis and Choledocholithiasis: a Feasibility Study

Md Ibrarullah · Tapas Mishra · Ambika P. Dash · Devanand Mohapatra · M. S. Modi

Received: 17 December 2012 / Accepted: 13 June 2013 / Published online: 22 June 2013

Abstract 'Single-sitting' laparoscopic cholecystectomy followed by endoscopic common bile duct clearance is emerging as a viable option for management of cholelithiasis and concomitant choledocholithiasis. The only disadvantage of the procedure is logistical since it requires co-ordination between two teams—the surgeons and the endoscopists. This limitation can be overcome in centres where both the procedures are performed by one team. With a considerable experience in endoscopy, we conducted a prospective study in a select group of patients to assess the feasibility of this single-sitting approach. The study included 38 patients with a radiological diagnosis of choledocholithiasis or jaundice at presentation. After laparoscopic cholecystectomy, the patients were turned prone and subjected to endoscopic retrograde cholangiogram, sphincterotomy and extraction of the common bile duct stone. The procedure was successful in 33 (87 %) of patients. The mean procedure time and hospital stay were 2 h, 20 min and 2 days, respectively. None of the patients had any major complications. We conclude that in a select group of patients, single-sitting laparoscopic cholecystectomy followed by endoscopic clearance of the common bile duct stone is safe and effective.

Keywords Choledocholithiasis · CBD Stone · Laparoscopic Cholecystectomy

M. Ibrarullah (✉) · M. S. Modi
Department of Surgical Gastroenterology, Apollo Hospitals, Bhubaneswar 751005, Odisha, India
e-mail: ibrarullahm60@gmail.com

T. Mishra · A. P. Dash
Department of Surgery, Hitech Medical College, Bhubaneswar 751023, Odisha, India

D. Mohapatra
Department of Medicine, Hitech Medical College, Bhubaneswar 751023, Odisha, India

Common bile duct (CBD) stone occurs in about 3–15 % of patients undergoing laparoscopic cholecystectomy (LC) [1–3]. In the era of open cholecystectomy, preoperative endoscopic clearance of CBD followed by open cholecystectomy was comparable [4–6] or even considered inferior [7] to single-stage cholecystectomy and choledocholithotomy. With minimally invasive surgery becoming the 'gold standard,' the endoscopic approach has seen a revival and has been considered a viable alternative to laparoscopic CBD exploration for concomitant CBD stone. Traditionally, gallstones with CBD stones are managed in two sittings i.e., endoscopic clearance of CBD stone followed by laparoscopic cholecystectomy or vice versa [8]. This entails involvement of two teams led by the endoscopist and surgeon for CBD stones and gallstones, respectively. The endoscopic approach has been found to be similar to laparoscopic CBD exploration in terms of successful stone clearance, morbidity, and mortality except for longer hospital stay [9, 10]. The latter disadvantage can be countered by carrying out the endoscopic CBD stone clearance as a single-sitting procedure in conjunction with LC [11]. The single-sitting approach, however, suffers from the logistic problem as it requires proper coordination between the surgeon and the endoscopist. We were encouraged to try this approach because of our considerable experience with both laparoscopy as well as endoscopy. We attempted to perform laparoscopic cholecystectomy as well as endoscopic clearance of the CBD stone ourselves without having the need to involve another team. In this report, we have presented our initial experience with this 'single-sitting' approach with respect to its feasibility and the outcome.

Material and Methods

All the patients with a diagnosis of cholelithiasis and choledocholithiasis, managed in the Department of Surgical

Gastroenterology between April 2009 and June 2012 were included in a prospectively maintained data base. Patients with radiological diagnosis of choledocholithiasis and/or those with jaundice at presentation were selected for the single-sitting approach. We excluded the following category of patients from the present trial—CBD stone diameter more than 1 cm, presence of cholangitis, co-morbidity restricting prolonged anaesthesia and any other contraindication for endoscopic retrograde cholangiopancreatography like duodenal obstruction, coagulopathy etc. Informed consent was obtained from the patients. Possibility of conversion to open surgery either due to failed laparoscopy or endoscopy was clearly explained to the patient. Patients with raised serum bilirubin received preoperative parenteral fluid, broad spectrum antibiotic and vitamin K. Laparoscopic cholecystectomy was performed in the conventional manner. A subhepatic tube drain was always placed to drain bile in the event of incomplete CBD clearance resulting in cystic duct blow out. After completion of cholecystectomy, the patient was turned prone and endoscopic retrograde cholangiogram (ERC) was performed followed by sphincterotomy and extraction of CBD stone by Dormia basket and/or balloon. Complete CBD clearance was confirmed by balloon sweeping and by cholangiogram. In case of any doubt, a 10 f–10 cm plastic stent was placed in the CBD that was removed 4 weeks later. In case of failed CBD clearance, the patient was turned supine and open choledocholithotomy and T-tube drainage was performed. In the postoperative period, the patients were assessed in the conventional manner for vital parameters, abdominal signs and for any LC- or ERC-related complications. Any unusual abdominal pain in the postoperative period was evaluated by serum amylase estimation to rule out pancreatitis. Oral fluids were resumed on the day of surgery and normal oral intake was restored by about 48 h. Patients were discharged as soon as they were ambulant and tolerated oral feeds.

Results

A total of 52 patients were admitted with a diagnosis of cholelithiasis and choledocholithiasis. Thirty-eight patients (6 males, 32 females) qualified for the single-sitting approach. The mean duration of symptoms was 21 weeks (2 days–12years). The clinical presentation in these patients has been listed (Table 1). Liver function test revealed mean serum bilirubin 3.5 mg/dl (range 0.9–6.7 mg/dl), SGOT 180 U/L (range 32–596), SGPT 190 U/L (range 17–472), serum alkaline phosphatase 438 U/L (range 95–1204). Deranged liver function i.e., raised serum bilirubin and/or liver enzymes were noted in 32 patients. Abdominal ultrasound examination showed gallstone in all, CBD stone in 23 and dilated CBD (Fig. 1) in 22 patients. MRCP was performed in 18 patients that confirmed CBD stone in all. The mean procedure time was 2 h 20 min (range 1 h 30 min– 2 h 45 min). The single-sitting procedure was successful in 33 patients (87 %) (Fig. 2). In five patients, the procedure failed leading to conversion to open surgery. This was during laparoscopic cholecystectomy stage in one patient and failure of ERC and CBD clearance in four patients. Difficult anatomy due to obliterated Calot's triangle was the cause of conversion during laparoscopy. ERC failure was due to cannulation failure, Dormia impaction, failed stone retrieval and impression of cystic duct stone in one patient each. The cystic duct stone, however, turned out to be a false impression on cholangiogram due to air bubble (Fig. 3). Eight patients in the immediate postoperative period had self-limiting moderate abdominal pain, but normal levels of serum amylase. None of the patients had any postoperative complications. The mean postoperative stay in patients who had successful single-sitting procedure was 2 days (range 1–3 days).

Table 1 Clinical presentation of the study group (n=38)

Pain	37
Fever	11
Jaundice	22
Acute pancreatitis	4
History of pancreatitis	1
Acute cholecystitis	2

Discussion

The single-sitting approach in the present study was applied to a select group of patients i.e., with radiological evidence of CBD stone or jaundice. Those with large-sized stone or co-

Fig. 1 Abdominal ultrasound showing dilated CBD and stone at the lower end

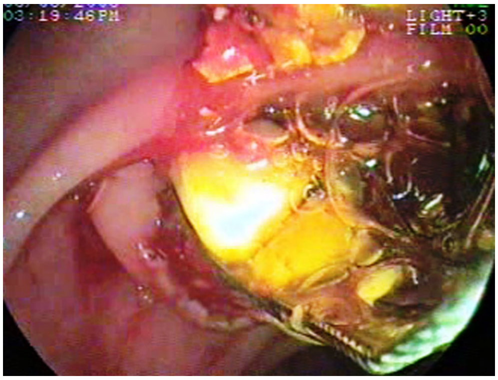

Fig. 2 Intraoperative endoscopic Dormia extraction of CBD stone

morbidity were excluded. Since the present study was carried out as a feasibility study, the strict inclusion or exclusion criteria was to ensure that there were not so many negative or failed endoscopic CBD interventions. Encouraged by the high success rate, we feel the scope of this approach can be extended to all except those in whom ERC is otherwise contraindicated [12].

Laparoscopic CBD exploration is the latest among all modalities of treatment for CBD stones. It is technically demanding requiring special instruments as well as skills thereby limiting its acceptance amongst surgeons. On the other hand, techniques of endoscopic clearance of CBD are more than three decades old and the expertise is widely available across the globe. In a multicenter prospective randomised trial, published more than a decade ago, Cuschieri et al. demonstrated that preoperative endoscopic clearance of CBD followed by laparoscopic cholecystectomy was comparable to one-stage laparoscopic cholecystectomy and CBD exploration in terms of success rate, morbidity and mortality [9]. This has further been proved in subsequent studies [13, 14]. The hospital stay and the total treatment cost were, however, significantly less with the one-stage laparoscopy group prompting the authors to recommend the latter as the procedure of choice [9, 10]. It is obvious that longer hospital stay, as well the resultant increase in the treatment cost, is the fall out of the staged procedure. Both of these could be addressed by performing endoscopic CBD clearance concurrently with cholecystectomy.

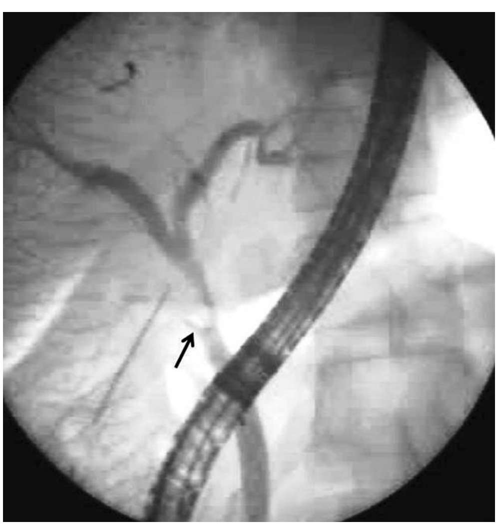

Fig. 3 Endoscopic retrograde cholangiogram showing filling defect (*arrow*) in the cystic duct that was misinterpreted as stone. At laparotomy this was found to be an artefact due to air bubble

In 1993, Deslandres et al. was the first to report laparoscopic cholecystectomy and concurrent endoscopic clearance of CBD [15]. In the technique reported by Deslandres et al., the laparoscopic surgeon passed guide-wire through the cystic duct. Emergence of guide-wire through papilla of Vater facilitated CBD cannulation, sphincterotomy and stone extraction by the endoscopist. This was followed by completion of cholecystectomy. This single-sitting approach was subsequently termed as ‘rendezvous’ technique [16]. We followed a slightly different technique similar to what has been reported by ElGeidie et al. [17]. The CBD stone removal was performed after cholecystectomy by turning the patient to prone position as is done normally for routine ERC. We concur with these authors [17] that passing the guide-wire through cystic duct offers no added advantage and the bowel distention resulting from ERC renders the subsequent steps of cholecystectomy cumbersome. In a review of 27 published studies on the rendezvous technique comprising of 795 patients, Greca et al. reported a success rate of 92.3 %, morbidity 5.1 % and the mortality rate 0.37 %. The mean procedure time was 104 min and hospital stay was 3.9 days. This compares favourably with our experience–success rate 87 %, no morbidity or mortality, mean procedure time 140 min and hospital stay 2 days.

In most of the centres, the endoscopic team is different from the surgical team. Therefore, preoperative clearance is likely to have logistical advantage as both the teams are allowed to carry out their respective jobs independent of each other. Failed cases of preoperative CBD clearance can be considered for open cholecystectomy and CBD exploration right away; on the other hand, when intraoperative CBD clearance fails, the advantages of laparoscopic cholecystectomy is lost because of conversion to laparotomy at this stage for CBD exploration. Such problem, however, can be minimised by careful selection of patient as was done in the present study. For our patients, we acted both as surgeons as well as endoscopists. Therefore, the logistical issue of co-ordination between two teams did not arise. This advantage has been highlighted by surgeons who are also trained to perform endoscopic clearance [17]. In a meta-analysis, Gurusamy et al. reported intraoperative CBD clearance to be as safe and effective as preoperative clearance with an added advantage of shorter hospital stay [18].

In conclusion, our study shows that in a select group of patients, single-sitting laparoscopic cholecystectomy followed by endoscopic clearance of CBD is safe and effective. The logistic problem of co-ordination between the surgical and the endoscopy team can be overcome if the surgeon has adequate expertise to perform endoscopy.

References

1. Sarli L, Iusco DR, Roncoroni L (2003) Preoperative endoscopic sphincterotomy and laparoscopic cholecystectomy for the management of cholecystocholedocholithiasis: 10-year experience. World J Surg 27:180–186
2. Schirmer BD, Winster KI, Edlich RF (2005) Cholelithiasis and cholecystitis. J Long-Term Eff Med Implants 15:329–338
3. Riciardi R, Islam S, Cante JJ et al (2003) Effectiveness and long term results of laparoscopic common bile duct exploration. Surg Endosc 17:19–22
4. Neoptolemos JP, Carr-Locke DL, Fossard DP (1987) Prospective randomized study of preoperative endoscopic sphincterotomy versus surgery alone for common bile duct stones. Br J Surg 294:470–474
5. Stain SC, Cohen H, Tsuishoysha M et al (1991) Choledocholithiasis. Endoscopic sphincterotomy or common bile duct exploration. Ann Surg 213:627–633
6. Stiegmann GV, Goff JS, Monsour A et al (1992) Precholecystectomy endoscopic cholangiography and stone removal is not superior to cholecystectomy, cholangiography and common duct exploration. Am J Surg 163:227–230
7. Kapoor R, Kaushik SP, Saraswat VA et al (1996) Prospective randomized trial comparing endoscopic sphincterotomy followed by surgery with surgery alone in good risk patients with choledocholithiasis. HPB Surg 9:145–148
8. Clayton ESJ, Connor S, Alexakis N, Leandros E (2006) Meta-analysis of endoscopy and surgery versus surgery alone for common bile duct stones with gallbladder in situ. Br J Surg 93:1185–1191
9. Cuschieri A, Lezoche E, Morino M et al (1999) E.A.E.S multicenter prospective randomized trial comparing two-stage vs single-stage management of patients with gallstone disease and ductal calculi. Surg Endosc 13:952–957
10. Rogers SJ, Cello JP, Horn JK et al (2010) Prospective randomized trial of LC + LCBDE vs ERCP/S + LC for common bile duct stone disease. Arch Surg 145:28–35
11. Greca GL, Barbagallo F, Sofia M, Latteri S, Russello D (2010) Simultaneous laparoendoscopic rendezvous for the treatment of cholecystocholedocholithiasis. Surg Endosc 24:769–780
12. Williams EJ, Green J, Beckingham I, Parks R, Martin D, Lombard M (2008) Guidelines on the management of common bile duct stones (CBDS). Gut 57:1004–1021
13. NIH state-of-the-science statement on endoscopic retrograde cholangiopancreatography (ERCP) for diagnosis and therapy (2002) NIH Consens State Sci Statements 19:1–26.
14. Martin DJ, Vernon DR, Toouli J (2006) Surgical versus endoscopic treatment of bile duct stones. Cochrane Database Syst Rev 19:CD003327
15. Deslandres E, Gagner M, Pomp A, Rheault M et al (1993) Intraoperative endoscopic sphincterotomy for common bile duct stones during laparoscopic cholecystectomy. Gastrointest Endosc 39:54–58
16. Miscusi G, Gasparrini M, Petruzziello L, Taglienti D, Onorato M, Otti M, Montori J (1997) Endolaparoscopic "Rendez-vous" in the treatment of cholecystocholedochal calculosis. G Chir 18:655–657
17. ElGeidie AA, ElEbidyGK NYM (2011) Preoperative versus intraoperative endoscopic sphincterotomy for management of common bile duct stones. Surg Endosc 25:1230–1237
18. Gurusamy K, Sahay KS, Burroughs AK, Davidson BR (2011) Systematic review and meta-analysis of intraoperative versus preoperative endoscopic sphincterotomy in patients with gallbladder and suspected common bile duct stones. Br J Surg 98:908–916